Computational Neuroscience — Module 03: Perception — Study Questions

1. What are the two streams of information in predictive coding? What direction does each flow?
2. Why is predictive coding efficient? What information does not need to travel upward?
3. Describe the canonical microcircuit. Which cortical layers encode predictions? Which encode errors?
4. What is layer 4's role in the canonical microcircuit?
5. What is the mismatch negativity (MMN)? How is it produced experimentally?
6. Why is the MMN considered evidence for predictive coding?
7. What does it mean that the MMN occurs even during sleep?
8. How is precision implemented at the neural level? What is gain control?
9. What does it mean to "turn up the gain" on a neural population?
10. Which neuromodulators control gain in sensory cortex vs. subcortical circuits?
11. Describe the visual processing hierarchy (V1 → V2 → V4 → IT). What does each area process?
12. How does predictive coding explain why you notice motion but not static features in your visual field?
13. What is the role of the thalamus in delivering sensory data to the cortex?
14. How might predictive coding explain the phenomenon of "seeing what you expect to see"?
15. What would happen if top-down predictions were removed? How would perception change?
16. How does attention interact with predictive coding? What does "paying attention" mean neurally?
17. Can predictive coding explain auditory perception (e.g., speech understanding in noise)?
18. How might hallucinations be explained by overactive top-down predictions?
19. What is repetition suppression and how does predictive coding explain it?
20. How does the computational neuroscience view of perception complement the cognitive science view (Unit 1, Module 03)?